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TPC, Where Art Thou?

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Abstract
Performance has and still is perceived by customers as one of the main differentiators of computer systems. Hence, having access to widely accepted, objective and verifiable performance data is paramount for customers in making informed buying decisions. The last ten years have seen the rise of many new DBMSs. With them came a flurry of benchmark definitions. Even if technically sound, these benchmarks alone cannot achieve the goal of delivering widely accepted, objective and verifiable performance data. In addition they need to be agreed upon by a large body of vendors, and they need an independent oversight authority that assures correctness of their results. There are two major standard organizations that work on achieving these goals: Standard Performance Evaluation Corporation (SPEC) and Transaction Processing Performance Council (TPC). This article centers around the TPC, its history, challenges and steps it is taking to stay relevant in the years ahead.

Keywords TPC · Benchmarks · Performance Analysis

1 How It All Started

It seems an eternity ago when the majority of business data processing was done in refrigerator sized computers and stored on disks the size of wedding cakes with the storage capacities like those of today's L3 caches . In those days Database Management Systems (DBMS) were mostly used for batch processing and predominantly implemented using IBM hardware and its hierarchical DBMS.

With the emergence of alternatives to IBM's mainframe, such as computer systems from Digital Equipment Corporation (DEC), Control Data Corporation (CDC), amongst others, running independent operating system, such as UNIX and independent DBMS, such as relational DBMS, IBM lost its monopoly on both hardware and software. Early performance claims were still based on TP1, a benchmark originally developed by IBM. TP1 batch processed Automated Teller Machine (ATM) transactions without taking network or user interaction into consideration.

In general, competition is good for customers, but in this case, it led to unfair marketing practices. This phenomenon, also referred to as "benchmarketing", is a marketing practice in which organizations claim superior performance based on non-standard benchmarks. The goal of running these tailored benchmarks is, simply put, to make one specific vendor's DBMS Solution[1] appear to be superior to that of its competitors by focusing on areas where it performs best. Tailored benchmarks included workloads with no specific execution rules, no minimum requirements on DBMS Solutions, no restrictions on how to price the DBMS Solution and, lastly, no obligations to document the benchmark executions including verification of their correctness.

Consequently, vendors published performance numbers using their own benchmarks and modified versions of their competitors benchmarks on system configurations that may not be deployed in real life, e.g., due to the lack of data or power supply redundancy, at prices at which no customers would be able to purchase them. In many cases it was impossible for potential customers to identify the best DBMS Solution for their needs.

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[1] A DBMS Solution includes both hardware (central computer, network and any peripheral hardware, etc.) and software (Operating System (OS), DBMS, Transaction Processing (TP) monitor middleware, etc. A DBMS Solution vendor provides both hardware and software – in some cases partnering with software vendors.

The DebitCredit benchmark that Jim Gray, together with 24 others from academy and industry, developed, fixed the technical shortcomings of TP1 by mimicking a realistic online transaction processing systems (OLTP), including network and user interactions. However, the "benchmarketing" issue was addressed only with the introduction of TPC-A. The TPC, founded in 1988, transformed DebitCredit into TPC-A, which was the first industry standard benchmark for transaction processing. It mandated production-oriented requirements, such as preventing the reporting of unsustainable peak performance numbers, requiring ACID compliance, and, most importantly, requiring the disclosure of the actual benchmark implementation and all test output as part of a Full Disclosure Report (FDR). In the years following, TPC-A evolved into TPC-B and eventually into TPC-C, which is still an active benchmark today. For detailed information about the beginning years of the TPC, please see [1].

The beginning years of the TPC were dominated by vendors that focused either on producing hardware or developing DBMS software. Vendors that produced both hardware and DBMS were rare, e.g., IBM and, to some degree, Teradata. Consequently, hardware vendors needed to team up with DBMS vendors to publish benchmark results. Short hardware generation cycles and cooperation of hardware vendors with multiple DBMS vendors resulted in a flurry of benchmark results. Publication of performance results sky rocketed at that time.

The remainder of this paper is structured as follows. After the TPC developed OLTP benchmarks it started exploring other benchmark domains. Sect. 2 provides a chronological overview of all TPC benchmarks that followed the footsteps of TPC-C. Constantly developing technology allowed more business processes to use computer systems in more complex ways. Sect. 3 explains how the TPC reacted to the changes by overhauling existing benchmarks. Sect. 4 describes the need for a new type of benchmark, named Express Benchmarks. While the number of TPC benchmark publications stagnated, publication of benchmarks derived from TPC benchmarks thrived. Sect. 5 gives examples of TPC derived benchmark numbers and how the TPC dealt with this threat. Sect. 6 illustrates the effects of industry consolidation. Sect. 7 lists efforts by the TPC to stay relevant in today's changing technology landscape while Sect. 8 concludes the paper.

2 Exploring New Benchmark Domains

After covering the OLTP benchmark needs of DBMS Solution vendors and customers the TPC ventured into other benchmark domains, with different degree of success, always learning valuable lessons along the way. Fig. 1 shows the timeline of when TPC benchmarks were active, i.e., when they were established and when they were retired. In 1994 the TPC presented its first decision support benchmark, TPC-D. Spurring the development of both hardware and software for five years, TPC-D was retired in 1999. The introduction of materialized views gave some database vendors a disproportionally large performance advantage, essentially breaking the performance metric of TPC-D.

TPC-D was subsequently split into two separate benchmarks, the reporting decision support benchmark, TPC-R, that allowed the use of sophisticated auxiliary data structures, whereby some of the query work was done during database load rather than during the performance test of its query workload, and TPC-H for an ad-hoc decision support workload [2]. TPC-R died a slow death due to the lack of benchmark results. It was retired in 2005, while TPC-H is still going strong as an active benchmark with 14 results in 2021, 3 results in 2020 and 8 results in 2019.

Two lessons were learned from these benchmarks. Firstly, the performance metric is the most vulnerable part of a benchmark, as vendors will always optimize for the largest performance benefits. Before settling on a metric for a benchmark it needs to be analyzed thoroughly to

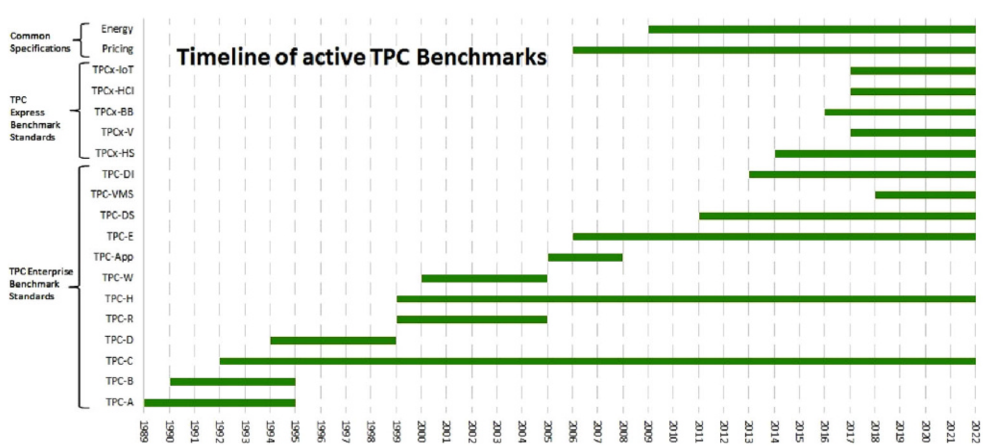

Fig. 1 History of TPC Benchmarks Between the Years 1989 and 2022

minimize the risk of vendors taking unfair advantage of it. Secondly, hardware vendors are not interested in benchmarks that showcase the avoidance of work, e.g., in case of materialized views.

With the emergence of electronic commerce, the TPC established TPC-W [2] in 2002. With minor changes it was later rebranded as TPC-App. It was an application server benchmark simulating the activities of a business-to-business transactional application server operating in a 24x7 environment. The workload exercised commercially available application server products, messaging products, and to some degree databases products. TPC-App never generated many benchmark publications, mostly because it did not exercise the back-end system running the DBMS. It was retired in 2008.

3 The Big Overhaul

In the years following, the TPC directed its effort towards developing modern versions of its two most successful benchmarks, TPC-C and TPC-H, and towards standardizing how systems are priced and energy is measured during benchmark execution. TPC-E [1] was introduced to address the shortcomings of TPC-C. Compared to TPC-C, TPC-E runs a more complex workload, requires RAID-protected storage, generates far less I/Os, and is much cheaper and easier to set up, run, and audit. Since its establishment in 2006 TPC-E produced 99 benchmark results.

TPC-DS [2] was established in 2011. It modernized the basic workload elements of TPC-H, database load, single-user test, multi-user test, and data maintenance test. It uses multiple interconnected star schemas (snowstorm schema), real world data, and 99 queries instead of 22. TPC-DS integrated ad-hoc and reporting queries into the same benchmark. The first version of TPC-DS required ACID compliance and a rigorous data maintenance test. Beginning with the second version, ACID was relaxed to a Data Accessibility Property (DAP) and the data maintenance test was reduced to inserts and deletes from fact tables. The DAP is satisfied if the DBMS is able "... to maintain operations with full data access during and after the permanent irrecoverable failure of any single Durable Medium containing tables, EADS, or metadata." [3, 4].

One of the trademarks of TPC benchmarks is their realistic pricing, i.e. that hardware and software used in benchmark results are priced at costs customers would be able to purchase them [5]. Until 2005, each TPC benchmark defined its own pricing requirements. The TPC, realizing that most of the pricing requirements were identical across benchmark definitions, developed a pricing document that could be referenced by all current and future benchmarks. This gave birth to a new benchmark specification category, named "Common Benchmarks". The pricing document was named TPC-Pricing [6, 7]. It was established in February of 2005.

Following the successful example of TPC-Pricing the TPC developed TPC-Energy [8, 9]. Like TPC-Pricing, TPC-Energy is a Common Benchmark. It measures the electrical energy required for the execution of a benchmark. As part of the TPC-Energy benchmark, the TPC provides a Energy Measurement System (EMS), which is a software package to facilitate the implementation of the TPC Energy Specification. Services provided by EMS are

1. power instrumentation interfacing,
2. power and temperature logging, and
3. report generation.

TPC-Energy was also the first collaboration between TPC and Standard Performance Evaluation Corporation (SPEC) as TPC-Energy licenses SPEC's implementation of a Power Temperature Daemon.

4 Third Wave – Express Benchmarks

Until the year 2017 all TPC benchmarks had been defined as paper specifications. A paper specification for a benchmark is a set of rules (clauses) that define how to run the benchmark. The TPC refers to a benchmark based on a paper specification as an enterprise benchmark. Because an enterprise benchmark does not provide an actual implementation, it is technology agnostic, thereby, leaving the actual benchmark implementation to the benchmark sponsor. This implementation is also referred to as a benchmark kit. It also allows using technology that had not been anticipated at the time the benchmark was defined.

4.1 Enterprise Benchmarks

There are, however, disadvantages of enterprise benchmarks. The bar for a new DBMS to enter the benchmark stage is very high for enterprise benchmarks, because the first benchmark publication of a DBMS requires the fully certified implementation of its benchmark kit. The certification process is done by a TPC certified auditor. Another disadvantage for hardware vendor that do not offer their own DBMS product is that they are required to team up with a DBMS vendor. There are not many DBMS vendors without hardware ties left in the TPC. Hence, pure hardware vendors have a difficult time finding partners with whom to publish performance numbers. After all, hardware vendors that have their own DBMS usually do not want to make another vendor's DBMS look better than their own. Finally, developing and approving an enterprise benchmark takes very long. Development takes long, because of the complexity in defining and agreeing on specification wording. Additionally, approval of a new enterprise benchmark requires a mail ballot, which by itself is a multi-step process. First step is for member companies to vote with simple majority for mail ballot, which can only take place during one of the 5 yearly meetings. Mail ballot can take up to 60 days and requires participation of at least 2/3 of all member companies. Approval of a new benchmark requires super-majority (2/3 of ballots). The solution was to allow benchmark specifications based on fully implemented, executable kits that do not require mail ballot.

4.2 Express Benchmarks

The TPC defined the framework for these kind of benchmark and named it express benchmark [10]. The cornerstones of express benchmarks are:

1. Fully implemented benchmark kits that execute the entire benchmark and that are easy to install and run
2. Peer review of benchmark results, and
3. Most new revisions of an express benchmark are approved during TPC Council meetings, which take place every 2 month instead of the otherwise required mail ballot for enterprise benchmarks.

4.2.1 TPCx-HS, a Benchmark for the Hadoop Ecosystem

The first TPC express benchmark, TPCx-HS [11], was established in 2014. It addresses the fast moving market of the Hadoop Ecosystem. TPCx-HS is based upon TeraSort (including TeraGen and TeraValidate). It uses regular map-reduce sort except for a custom partitioner that splits the mapper output. Version 2 of TPCx-HS V2 includes support for Apache Spark, MapReduce (MR2) and support for on-prem publications as well as in cloud deployments.

4.2.2 TPCx-BB, a Benchmark for Big Data

The second TPC express benchmark, TPCx-BB [12], was established in 2016. It measures the performance of both hardware and software components of Hadoop-based Big Data systems. It addresses the 3 Vs in big data, volume, variety, and velocity. Volume is addressed by providing data generators that scale up to petabytes of data. Variety is addressed by using data in form of structure data, semi-structured data, and un-structured data. The structured data and SQL queries are based on TPC-DS. Semi-structured data is implemented as web logs while un-structured data is implemented with product reviews. Both semi- and un-structured data is queried by machine learning algorithms, user defined functions, and procedural programs. Velocity is implemented by periodic refreshes of the data repository.

4.2.3 TPCx-V and TPCx-HCI, two Virtualization Benchmarks

In 2017 the TPC released two virtualization benchmarks, TPCx-V [13] and TPCx-HCI [14]. TPCx-V measures the performance of a virtualized server running a mixed database workload comprising of OLTP and DSS workloads. It stresses many cloud services, such as multiple virtual machines (VM) running at different load demand levels, and large fluctuations in the load level of each VM. TPCx-HCI measures the performance of Hyper-Converged Infrastructure clusters under a demanding database workload. TPCx-HCI extends TPC-V for virtualized hardware and software of converged storage, networking, and compute resources of an HCI platform.

4.2.4 TPCx-IoT, a Benchmark for Internet of Things

In 2017 the TPC also released its first benchmark for the Internet of Things (IoT) market, TPCx-IoT [15]. It measures the performance of software and hardware solutions for IoT gateways running on commercially available hardware- and software-platforms. These systems usually ingest massive amounts of data from a very large number of devices, while providing real-time analytic functions on that data. TPCx-IoT's workload generator is based on the Yahoo! Cloud Serving Benchmark framework (YCSB) [16]. YCSB was used to develop TPCx-IoT because of it being an easy to adapt open source tool and for its built-in database interface layer, which allows for connectivity to many common open-source database management systems including NoSQL DBMS.

It is not uncommon for enterprise benchmarks to take more than five years to be developed. Express benchmarks have a much shorter development time. For instance, TPCx-IoT took just over 2 years to be developed (4/23/2015 to

Table 1 Express Benchmark First Results

Benchmark	Establishment Dates	Number of Result	Adaptation [years]
TPCx-HS	6/17/2014	32	3.6
TPCx-V	11/12/2015	5	3.8
TPCx-BB	2/19/2016	21	0.9
TPCx-VCI	1/19/2017	1	4.9
TPCx-IoT	6/7/2017	11	1.5

6/7/2017), TPCx-HS took only five months to be developed (2/6/14 to 6/17/14). The main reason for the short development time for these benchmarks is that they their kits are based on existing frameworks and their specifications are short documents.

Table 1 shows when each of the express benchmarks were established, how many benchmark results are currently on the book for each of the benchmarks and how many years it took to develop the benchmark.

5 Relevance of Benchmarks

The current rate of benchmark result publications is still lacking far behind the numbers we had seen in the 1990s to early 2000s. Is performance not a major concern for customers anymore when comparing different DBMS Solutions? Looking at the large amount of marketing material that is using performance numbers it quickly becomes obvious that performance is still paramount for advertising DBMS Solutions. If performance is still important to customers, then benchmarks are as well. Today's benchmark situation is very similar to that of the early 80s when companies engaged in "benchmarketing". While back then DBMS Solution companies used modified versions of TP1 and modified versions of their competitors' benchmarks, nowadays DBMS Solution companies use mostly benchmarks derived from TPC benchmarks to make their marketing claims.

5.1 Examples Of Workloads Derived from TPC Benchmarks

The following examples are just a few to demonstrate the extend to which benchmarks derived from TPC benchmarks are used in today's marketing. Qubole uses a subset of the TPC-DS queries to claim that Hive on Qubole runs 4x faster than Hive on Alternative Platforms[2].

AtScale uses a subset of the TPC-DS queries to claim that AtScale's Accelerator on top of Redshift speeds up query performance by a factor of 12.5 compared to Redshift alone[3].

YugabyteDB publishes TPC-C performance numbers at 10, 100, 1,000 and 10,000 warehouses[4].

Concurrency Laps publishes elapsed times of all 99 TPC-DS queries on STARBURST 350-E, EMR PRESTO 0.230, EMR SPARK 3.0.0 and EMR HIVE 3.1.2[5]. These are just a view examples of today's "benchmarketing".

None of the above performance claims are backed by published TPC performance numbers. Many are using only a subset of the TPC-DS queries, only include a subset of the actual performance tests of TPC-DS, which consist of a Load Test, a Single-User Test, a Multi-User Test and a Data Maintenance Test and do not necessarily adhere to the stringent system requirement rules of audited TPC-DS results, which can be found on the TPC results page[6].

5.2 Rules Under Which TPC Derived Benchmarks Can Be Used

The use of benchmarks derived from TPC benchmarks is allowed by the TPC as long as TPC policies [17] are not violated. When the TPC realized that there are an increasingly high number of incidents in which companies create benchmarks that look very similar to TPC benchmarks, and, in many cases, even refer to the TPC name of the benchmark, e.g. TPC-DS, the TPC included policy wording to regulate the use of TPC benchmarks. The center of these rules is the term "non-TPC Benchmark". The TPC policies Clause 0.2.36 states that a non-TPC Benchmark is "A derived implementation of a TPC Benchmark Standard that does not maintain all the requirements of the TPC Benchmark Standard from which it is derived." The use of a non-TPC Benchmark Standard is regulated in Claue 8.3 of the TPC policies, which states "... It is permissible to create a non-TPC Benchmark Standard. Any non-TPC Benchmark Standard must adhere to the following requirements: ... The Use of TPC Material (Policies § 8.1) and Fair Use Policy (Policies § 8.2.1) must not be violated. ... All deviations from the TPC Benchmark Standard in question must be explicitly noted. ... Results based on a non-TPC Benchmark must be clearly identified as not being comparable to an official TPC Benchmark Result."

Some DBMS solution vendors incorporate non-TPC benchmarks directly into their documentation. This includes schema definition, queries and sample data. Snowflake incorporates TPC-DS[7] and TPC-H [8]. Cockroach DB incorporates TPC-C into their documentation[9]. Amazon uses TPC-

[2] source: Hive on Qubole runs 4x faster than Hive on Alternative Platforms https://www.qubole.com/blog/hive-on-qubole-runs-4x-faster-than-alternatives.

[3] source:AtScale Cloud Data Warehouse Benchmark: Amazon Redshift https://www.atscale.com/wp-content/uploads/2019/11/ATS3346_Benchmark_Survey_Report_Redshift-updated_compressed.pdf.

[4] source: TPC-C Benchmark: 10,000 Warehouses on YugabyteDB https://www.yugabyte.com/blog/tpc-c-benchmark-10000-warehouses-on-yugabytedb.

[5] source: Querying 6.35 Billion Records – a TPC-DS Performance and Cost Comparison between Big Data platforms Starburst Enterprise and EMR SQL engines https://www.concurrencylabs.com/blog/starburst-enterprise-vs-aws-emr-sql.

[6] https://www.tpc.org/tpcds/results/tpcds_results5.asp?version=3.

[7] https://docs.snowflake.com/en/user-guide/sample-data-tpcds.html.

[8] https://docs.snowflake.com/en/user-guide/sample-data-tpch.html.

[9] https://www.cockroachlabs.com/docs/v22.1/performance-benchmarking-with-tpcc-large.

Adaptec	Information Decisions	Samsung Electronics
Advanced Logic Research	Informix Software	Sanford Bernstein
Amdahl	Intel Corporation.	Sarion Systems Research
BEA Systems	Intergraph	SCO
Bull S.A.	ITOM International	Sequent Computer
Compaq Computer Corp.	Microsoft Corporation.	Siemens Nixdorf
Computer Associates	Mitsubishi Electric	Information Systems
Data General Corp.	Motorola	Silicon Graphics
Debis Systemhaus	Morgan Stanley	Software AG
Dell Computer Corporation	Mylex	Stratus Computer
Digital Equipment Corp.	NCR	Sun Microsystems
EDS	NEC Systems Laboratory	Sybase
EMC Corp.	Nikkei Business Publications	Tandem Computers
Fujitsu/ICL	Novell	Toshiba Corp.
Hewlett-Packard	OKI Electric Industry	Tricord Systems Inc.
Hitachi SW	Olivetti S.p.A	Unisys Corporation
IBM Corp.	Oracle Corporation	University of Pavia
IDEAS International	Performance Tuning Corporation.	White Cross Systems

Fig. 2 TPC Member Companies January 1995

Actian	Hitatchi	Nettrix
Alibaba	Huawei	Nutanix
AMD	IBM	NVIDIA
Cisco	Inspur	Oracle
Dell	Intel	RedHat
Fujitsu	Lenovo	TTA
Hewlett Packard Enterprise	Microsoft	VMWARE

Fig. 3 TPC Member Companies July 1995

DS in their Amazon Athena TPC-DS Connector[10]. Incorporating TPC benchmarks directly into their documentation enables customers to run TPC workloads easily. However, one needs to understand that these results are not audited/certified, that they must be accompanied with statements clearly indicating that these results are not official TPC benchmark results, but those of derived benchmarks, whose performance cannot be compared with that of published/audited benchmarks.

6 Acquisition Phase

Membership in the TPC is renewed annually, but companies can join the TPC anytime with prorated dues. The number of member companies fluctuated over time. However, in the beginning of the new millennium a general downward trend in TPC membership number started. While in January 1995 the TPC had 54 member companies (see Fig. 2), that number reduced to 21 as of July 2022 (see Fig. 3).

As the computer industry matured, some companies merged, some that struggled to keep market-share were acquired by competitors. This was especially true within the TPC and caused the TPC membership to drop significantly. For instance, Digital Equipment Company was acquired in 1998 by Compaq. In 1999 Compaq acquired Tandem, while Compaq was subsequently merged with Hewlett-Packard. Oracle acquired BEA in 2008 and Sun in 2009. In 2004 VMware was acquired by EMC which was aquired by Dell Corporation in 2015.

This series of acquisitions not only significantly reduced the number of TPC member companies, but it also increased the number of hardware companies that had a DBMS with significant market share in their portfolio. Consequently, these companies focused on promoting their own database product(s) rather than collaborating with other database companies. This significantly reduced the number of benchmark publications.

7 Staying Relevant

In order to stay relevant the TPC has constantly adjusted to the changing market conditions. With the emergence of new DBMS Solutions the TPC noticed that, instead of publishing audited TPC benchmarks, these DBMS Solution companies use TPC derived benchmarks. Instead of disallowing the use of derived benchmarks, the TPC explicitly allows the use of them as described in the previous section. I consider the fact that companies use non-TPC benchmarks a tribute to the completeness, thoughtfulness and quality of the benchmarks developed by the TPC. One can only speculate why companies run non-TPC benchmarks rather than the real deal. One reason might be lack of functionality in the DBMS Solution. There are many TPC-DS claims using only a subset of the TPC-DS queries, which leads to believe those DBMS Solutions do not run the remaining queries well enough to be included in their claims. Or, some queries fail to run because the DBMS Solution does not support certain SQL constructs, such as ROLLUP. Other performance claims that are based on TPC-DS do not include all the execution phases of the original TPC-DS benchmark, such as the multi-user run or proof of Data Accessibility Property. While allowing the use of non-TPC benchmarks, the TPC emphasizes the fact that TPC benchmarks were developed as they are for a reason. Running a derived TPC benchmark by Cherry-Picking those parts one can run best distorts the performance measured by the derived benchmark. When customers realize this they are most likely to demand fully audited benchmark results.

7.1 Retrofitting TPC Benchmarks For The Cloud

Over recent years Cloud Computing has become more accepted and more common among customers. The TPC has made efforts facilitating the execution of TPC benchmarks in the cloud. In February 2017 the pricing specification

[10] https://docs.aws.amazon.com/athena/latest/ug/athena-prebuilt-data-connectors-tpcds.html.

added the term Licensed Compute Services. In clause 0.1.2 the pricing specification defines Licensed Compute Services as " ...Publicly offered processing, storage, network, and software services that are hosted on remote computer servers accessed via a Wide Area Network (e.g., the Internet), which are not located or installed on a Customer's premises. A Customer pays a license fee to the Licensed Compute Services vendor for the use of the processing, storage, network, and software services". Since then the pricing specification has undergone many cloud related revisions including, but not limited to Pay As You Go Pricing Methodology, Conversion Rate, Currency Conversion, Currency Conversion Source, NonLocal Currency, Price Performance, Priced Currency, Priced Locale, and Primary Currency. These changes were necessary so that benchmarks run in the cloud can be priced properly.

TPC-C, TPC-E, TPC-H and the first version of TPC-DS required DBMS Solutions to demonstrate that they are ACID[11] compliant. To demonstrate durability a DBMS Solution has to successfully run a test in which a permanent unrecoverable failure of a durable medium occurred, e.g. a hard disk. This durability test has to be performed by physically failing the durable media, e.g. pulling the disk. This is not necessarily feasible in the cloud. For TPC-H publications in the cloud the TPC allowed failing entire storage nodes instead of individual disks. For instance, see Alibaba's result using OceanBase[12].

7.2 TPC's Open Source Initiative

The increasing number of TPC benchmarks and the shrinking number of member companies reduces the ability of the TPC to develop new benchmarks and to make significant changes to existing benchmarks. Back in 1995 the ratio of number of TPC benchmark standards to member companies was about 1:25, i.e. on average 25 member companies maintained one TPC benchmark. Today this ratio is 11:21, i.e. on average 2 member companies maintain one TPC benchmark. In reality it is even worse as not all member companies contribute to the same extend. The introduction of express benchmarks alleviated this issue to some extend.

To further boost the TPC's ability to develop benchmarks and to make significant contributions to existing benchmarks, the TPC embarked on an Open Source Initiative (TPC-OSS). The TPC-OSS initiative leverages community driven development and testing resources, promotes TPC to a generation of engineers and developers that expect access to source code, gets the TPC brand out in the developer community more associated with easy to use database workloads and becomes more approachable to Academia, which currently views the TPC as rigid and closed off. As its first project the TPC-OSS collaborated with HammerDB starting in May 2019. HammerDB has been one of the first open source products to use TPC based benchmarks. As the first step towards bringing the TPC and HammerDB communities together the HammerDB source code was moved to the TPC-Council GitHub repository. Since then the total GitHub release downloads reached 158,197.

8 Conclusion

This article demonstrated that standard benchmarks developed by industry led consortia are still relevant. It gave insights into the reasons that started standard consortia like the Transaction Processing Performance Council (TPC). Because "Change is the only constant in life."[13] the TPC had to adapt to technological and economical changes in the tech sector. This article revealed how the TPC stayed relevant for 34 years by adapting to these changes. Over its lifetime, the TPC has been constantly exploring new benchmark domains, overhauling its existing benchmarks, developing innovative benchmark models and coping with a reduced membership. Against all odds, the TPC never lost sight of its main objective: *Delivering widely accepted, objective and verifiable performance data to the industry*.

Declarations

Conflict of interest The authors do not have any financial or non-financial interests that are directly or indirectly related to the work submitted for publication.

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